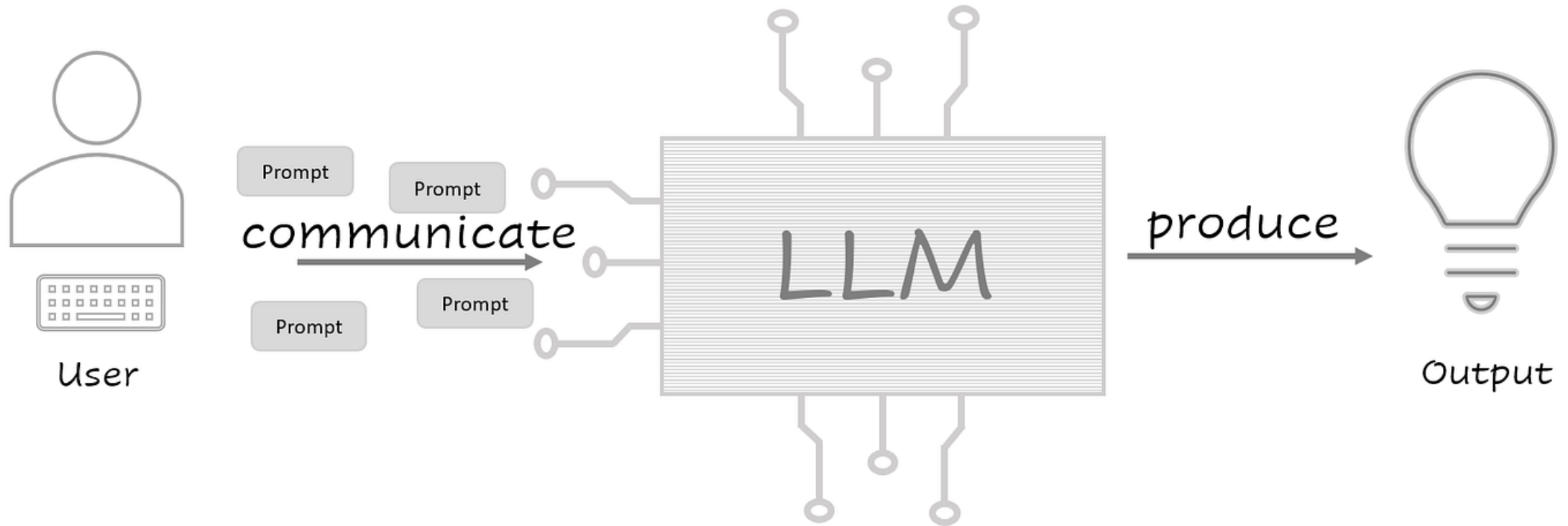


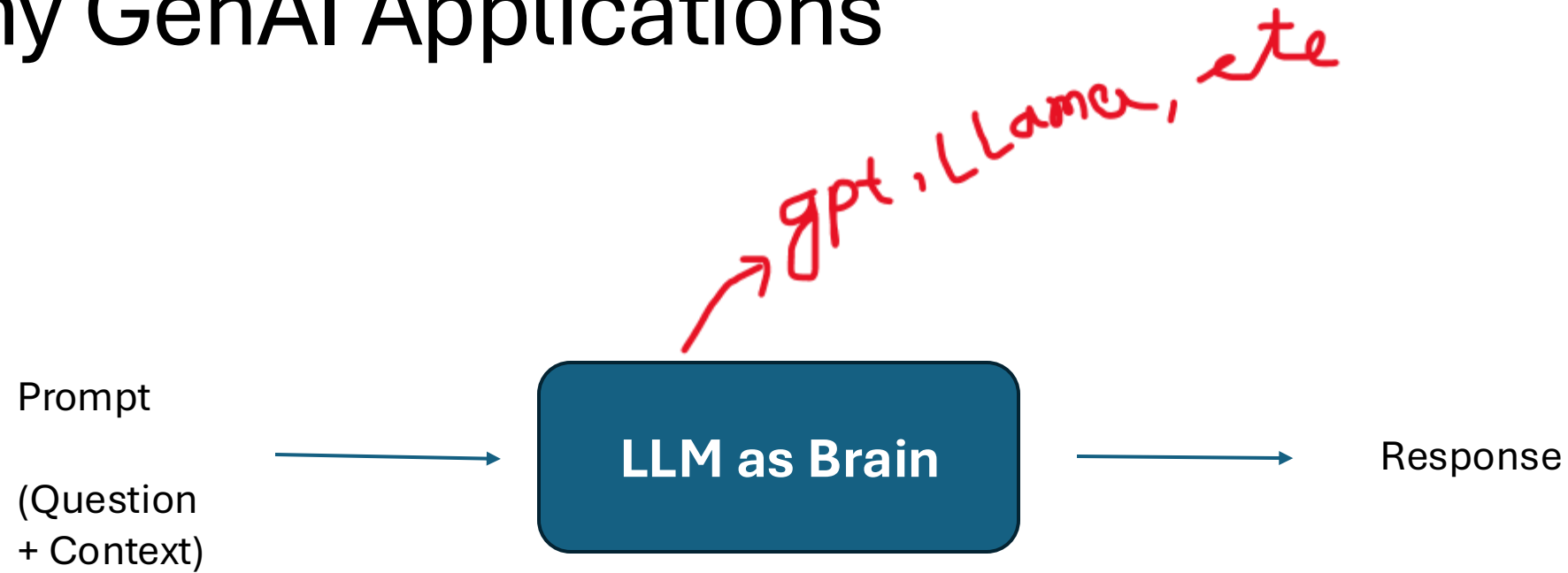
Prompt Engineering



Prompt Engineering

Design the Prompt, Define
the Output

Any GenAI Applications



Prompt Engineering

Definition: **Prompt engineering** is the process of developing, designing, and optimizing prompts to efficiently utilize language models (LMs) for various applications. It is considered both an art and a science because Large Language Model (LLM) outputs are non-deterministic.

Importance: It helps in understanding the **capabilities and limitations** of LLMs. Effective prompts guide models toward generating accurate, relevant, and safe responses.

Role in Generative AI: It acts as a **roadmap**, steering the AI toward a specific desired output. It allows developers to interface with LLMs and external tools while improving model safety.

Real-World Applications: Used in **natural language processing (NLP)** tasks like question answering, arithmetic reasoning, and building domain-specific capabilities

Components of an Effective Prompt

- An optimal prompt typically includes the following logical boundaries:
 - **Role (Identity):** Define the purpose and communication style (e.g., "You are a senior software engineer").
 - **Task (Instructions):** Explicit guidance on what the model should do and rules it must follow.
 - **Context:** Supporting information or proprietary data relevant to the specific request.
 - **Constraints:** Specific limitations, such as word counts or forbidden topics.
 - **Output Format:** Desired structure (e.g., Markdown, JSON, or a bulleted list).
 - **Examples:** Handfuls of input-output pairs to demonstrate the desired pattern

Bad Prompt vs Good Prompt

Aspect	Bad Prompt	Good Prompt
Specificity	Write something about climate change.	Write a persuasive essay arguing for stricter carbon emission regulations.
Constraint	Write a long poem.	Write a sonnet with 14 lines exploring themes of love and loss.
Logic	Create a marketing plan.	1. Identify the target audience. 2. Develop key messages. 3. Choose channels.

System vs. User Prompts

Developer (System) Messages: Provide high-level rules, business logic, and "instructions" that have higher authority.

User Messages: Provide the specific input or arguments to which the system rules are applied

Prompting Techniques

Zero-Shot Prompting



Definition: Providing instructions without any examples.

Use Case: Simple tasks like idea generation or basic translation.

Pros/Cons: Fast but may lack precision for complex tasks.

One-Shot & Few-Shot Prompting

Definition: Including one or more input-output examples in the prompt.

Example: Input: "Cat" Output: "Small mammal" | Input: "Dog" Output: "Canine" | Prompt: "Elephant".

Advantage: Steers the model toward a task without fine-tuning

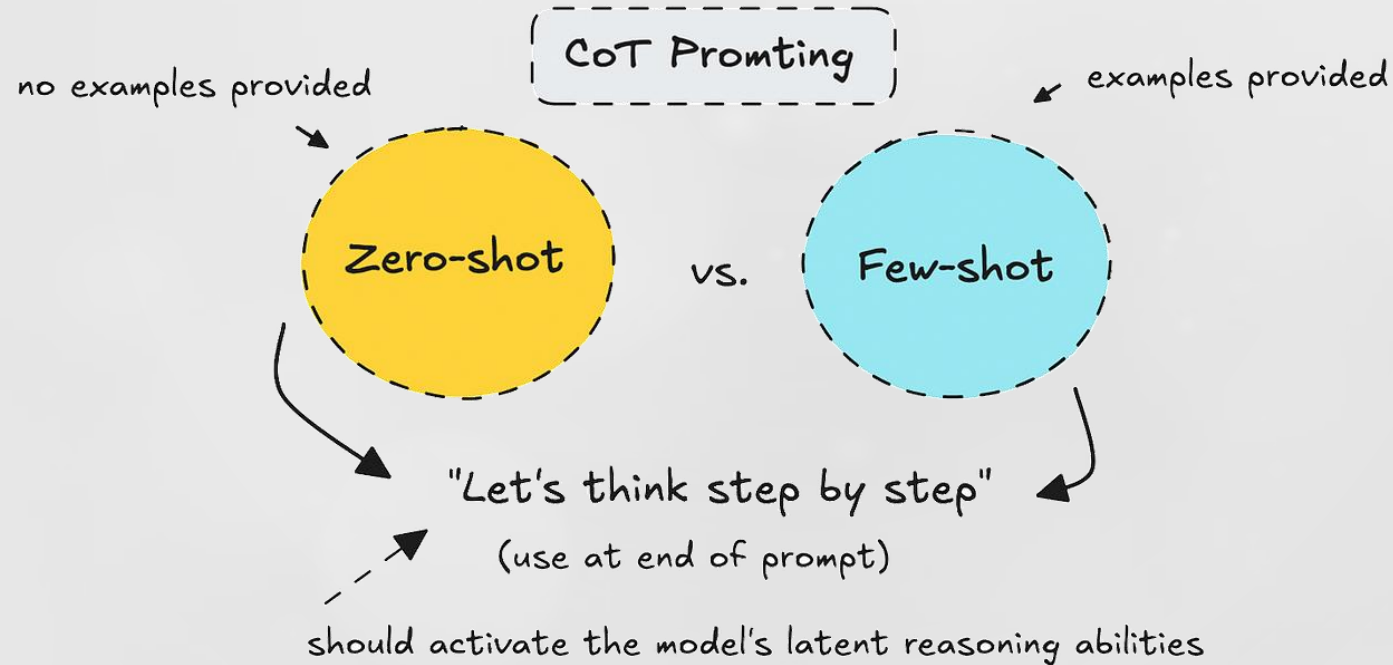
Prompting Techniques

Chain of Thought (CoT)

Definition: Encouraging the model to break down complex reasoning into intermediate steps.

Example Prompt: "Solve this problem step-by-step: John has 5 apples, he eats 2..."

Advantage: Leads to more comprehensive and well-structured final outputs.



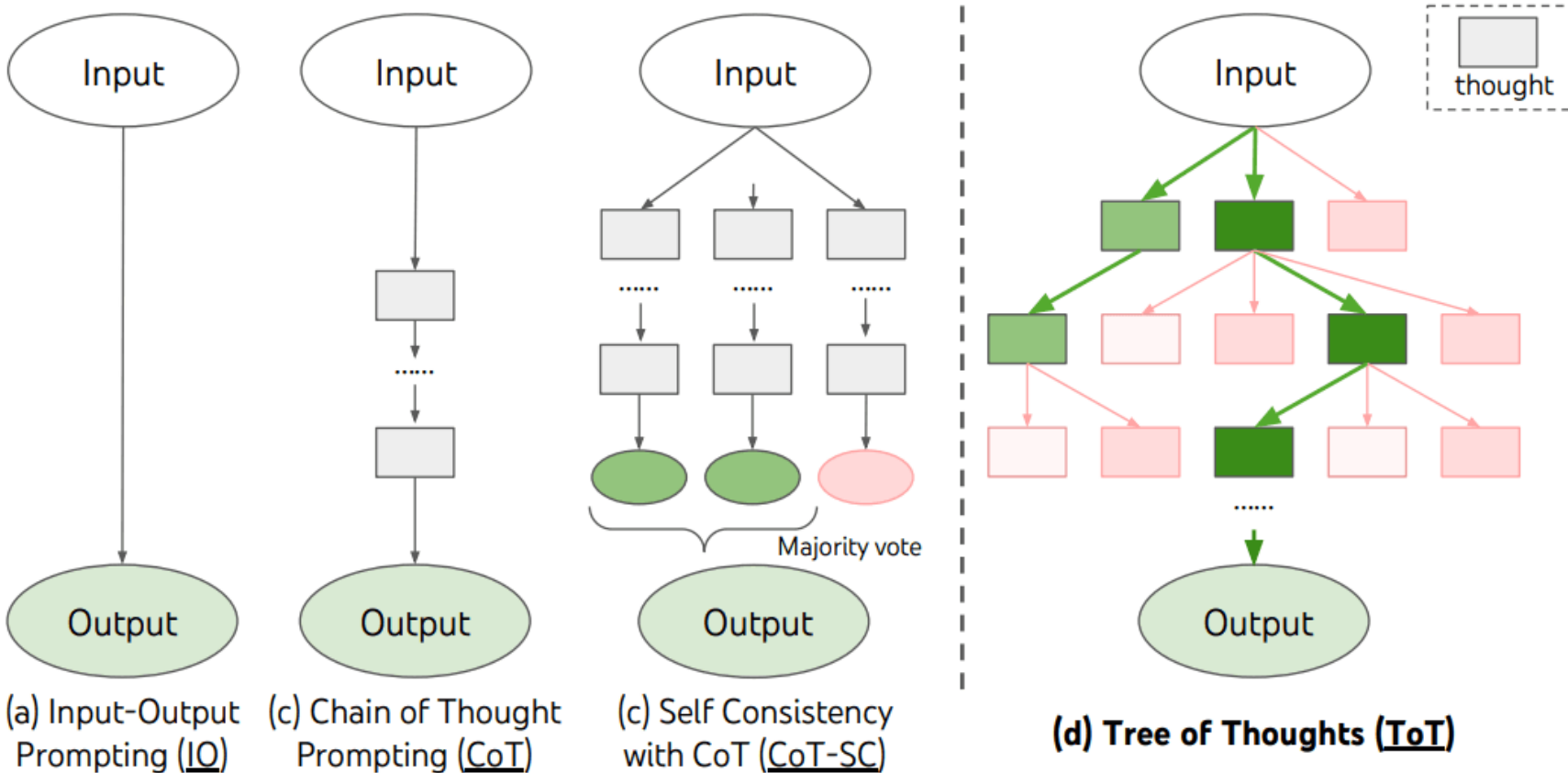
Prompting Techniques

Instruction Prompting

- **Instruction prompting** is a technique where task instructions are provided directly to a Large Language Model (LLM) instead of relying on demonstrations or examples. It serves as a more efficient alternative to few-shot prompting, which can be expensive in terms of token usage and is often restricted by the model's context length limits.
- **Key Characteristics and Mechanism**
 - **Instructed LMs:** This method utilizes models (such as **InstructGPT**) that have been specifically fine-tuned to understand and follow human intentions.
 - **Fine-tuning Process:** These models are trained using high-quality tuples consisting of a **task instruction, input, and ground truth output**. A common method for achieving this alignment is **Reinforcement Learning from Human Feedback (RLHF)**.
 - **Alignment:** The primary goal of instruction-based fine-tuning is to ensure the model is better aligned with human intention, which significantly reduces the "cost" or complexity of communication.

Prompting Techniques

Tree of Thought Prompting



Settings to keep in mind

- When prompting a language model, you should keep in mind a few settings.
- You can get very different results with prompts when using different settings.
- One important setting is controlling how deterministic the model is when generating completion for prompts.
- **Temperature** and **top_p** are two important parameters to keep in mind.
- Generally, keep these low if you are looking for exact answers. Keep them high if you are looking for more diverse responses

Settings to keep in mind

- **Temperature** - In short, the lower the temperature, the more deterministic the results in the sense that the highest probable next token is always picked. Increasing temperature could lead to more randomness, which encourages more diverse or creative outputs. You are essentially increasing the weights of the other possible tokens. In terms of application, you might want to use a lower temperature value for tasks like fact-based QA to encourage more factual and concise responses. For poem generation or other creative tasks, it might be beneficial to increase the temperature value.
- **Top P** - A sampling technique with temperature, called nucleus sampling, where you can control how deterministic the model is. If you are looking for exact and factual answers keep this low. If you are looking for more diverse responses, increase to a higher value. If you use Top P it means that only the tokens comprising the top_p probability mass are considered for responses, so a low top_p value selects the most confident responses. This means that a high top_p value will enable the model to look at more possible words, including less likely ones, leading to more diverse outputs.
- The general **recommendation is to alter temperature or Top P but not both.**
- **Max Length** - You can manage the number of tokens the model generates by adjusting the max length. Specifying a max length helps you prevent long or irrelevant responses and control costs.

Thank you

